Union-by-size selection and exact machine addition

Nine declarations and seven fixed controls compiled, with sixteen separate named axiom readbacks. Independent source-fidelity review is pending. No scientific run or timing sample was taken.

Construction. The model uses a fixed-length Array USize and the native <= comparison: ties select the first root as child. The selected parent and child indices are in bounds. The update changes only the parent size slot and retains the child and unrelated slots. For disjoint finite component sets, the two cardinalities sum to at most the node count. If stored root sizes equal these cardinalities and n<USize.size, parent addition is exact rather than modularly wrapped.

Composition. size_selected_union_preserves instantiates the preceding two-find/union theorem with the Boolean selected by these machine sizes. Either size ordering preserves canonical coefficients, canonical fixed anchors, decoded relation validity and existence of a forest rank under that theorem's premises. This attaches a source-style choice to the previously general orientation result.

Source boundary. The native SignedGraph.edge implementation chooses at size[ru]<=size[rv] and updates size[parent]+=size[child]. The proof binds its retained source bytes. Stored-size correspondence to actual graph-component cardinalities remains an explicit premise; initialization, path-compression preservation, component merger and node growth must still establish it inductively. The fixed-size model does not prove allocation, vector growth, operation-counter overflow or the native return path when both components have anchors. The incoming geometric relation remains a premise of the graph theorem.

Controls. Seven controls check tie-breaking, both size orderings, parent sum, stale child-slot retention, unrelated-slot preservation and wraparound at the machine maximum. These controls explicitly assume a 64-bit platform. The general exact-addition theorem instead uses symbolic USize.size. Early controls failed because kernel reduction cannot simply choose the platform width; the explicit conditional formulation preserves that distinction. All failed attempts and compiler warnings are retained.

Verification. Final source modules and readback exited zero with unchanged bound input files. The deliberate false theorem exited one. The inventory checker accepts all sixteen exact names, including the axiom-free bounds declaration, permits only propext, Classical.choice and Quot.sound otherwise, and rejects empty, wrong-name and sorryAx inventories. Imported compiled dependencies were reused; this is not a clean dependency rebuild or an independent binary recheck. Before/after hashes do not exclude transient concurrent edits.

Next step. Independently review the array and machine-word correspondence, then prove root-size/component-cardinality invariance through initialization, node insertion, path compression and union. Keep the two-anchor return constraint separate until composed. The diagram records proof dependencies, not measured performance. Frozen scientific scope, failure accounting and campaign limits remain unchanged.